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Breaking Gender Barriers: Experimental Evidence on Men in Pink-Collar Jobs[†]

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I investigate men's limited entry into female-dominated sectors through a large-scale field experiment. The design exogenously varies recruitment messages by showing photographs of current workers (male or female) and providing information on the share of workers who received high evaluations in the past (higher or lower). A male photograph has no impact on men's applications, but informing about a lower share of high evaluations encourages men to apply and enables the employer to hire and retain more talented men. The impact of this informational intervention remains positive for the employer also accounting for its effects on female applicants and hires. (*JEL* C93, D83, J16, J22, J23, J24, M51)

In the last half century, the shift from brawn- to brain-intensive occupations has eroded the traditional advantage that men have enjoyed in the labor market (Autor and Wasserman 2013). A declining manufacturing share of employment in the United States and other OECD countries has increased the risk of men's involuntary displacement, long spells of unemployment, and idleness (Ngai and Petrongolo 2017; OECD 2019). At the same time, the demand for female-dominated occupations, such as health care and education, has grown and often faces staffing shortages (Blau and Kahn 2017; Graves and Kuehn 2021). Despite this, male labor remains a relatively untapped resource for addressing these hiring challenges, and organizations are still grappling with how to attract more men into these growing pink-collar occupations.[1]

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[1] The term "pink-collar" is commonly used for care-oriented jobs and fields historically considered to be women's work. The scarcity of programs targeted at men's entry into female-dominated roles contrasts with the abundance of interventions that encourage women into male-dominated sectors (see, e.g., Koch, Irby, and Polnick 2014). Lessons from these experiences are difficult to apply to the context of this paper if men and women face different constraints and if perceptions or attitudes toward female- and male-dominated occupations differ.

The organizational problem is twofold. First, employers need to know how to increase male applications while maintaining cost-effectiveness. This requires understanding the barriers that deter men from supplying their labor to female-dominated occupations and knowing whether they can be addressed through cheap organizational changes. Second, employers must consider the potential trade-offs of policies that boost male entry. Such policies may attract men who are not well suited for the job or discourage qualified female applicants. Therefore, investigating who gets hired and their job performance is crucial to comprehensively evaluate the effectiveness of interventions that aim to increase male representation in female-dominated jobs.

In this paper, I address this dual organizational challenge through a large-scale natural field experiment, conducted in collaboration with a major UK social work employer. This partnership allows me access to data beyond the application stage, including hiring outcomes, on-the-job performance, and retention for both genders for over two years. The experimental design generates exogenous variation in the content of recruitment messages sent to potential applicants for the job offered by the partner organization. By varying these recruitment materials, I can identify supply-side barriers that prevent men from entering female-dominated jobs. Furthermore, this approach offers a practical and cost-effective policy that firms can readily adopt.[2]

Social work provides a good setting for investigating my research questions because it is a high-growth female-dominated occupation, with social work roles expected to grow twice as fast as other occupations in the United States (US Bureau of Labor Statistics 2019). However, the proportion of male social workers has remained unchanged since 1970 (Blau, Simpson, and Anderson 1998). Social work, like other caring professions, also faces challenges in attracting and retaining talent, particularly among men. In the United Kingdom, annual turnover rates have reached 17 percent, and 12 percent of social work positions remain unfilled (Skills for Care 2023). Discovering effective policies that attract talented men to social work will satisfy not only the need for diversity but also efficiency.

In designing the messages sent to potential applicants, I focus on two factors that may deter men from considering female-dominated jobs: gender identity and expected success on the job. Both dimensions were leading concerns for the recruitment personnel of the partner organization, who worried about talented men not perceiving social work as a suitable career choice.

Regarding gender identity, men may avoid social work due to concerns about it not being appropriate for their gender, even if they could be successful. Selecting into female jobs may threaten their masculinity (Akerlof and Kranton 2000, 2005; Baranov, De Haas, and Grosjean 2023) or internalized gender norms (Bursztyn and Jensen 2017; Oh 2023; Del Carpio and Guadalupe 2022). Additionally, they may have reservations about working in an environment where a majority of employees are women (Becker 1957) or believe that employers or clients prefer female workers (Folke and Rickne 2022; Booth and Leigh 2010; Rich 2014). Many

[2] The experiment was double-blinded to reduce experimenter demand and to not interfere with the hiring process. Compared to alternative variations (e.g., in incentives), the design preserves existing organizational systems and does not require hands-on administration, making it scalable (Al-Ubaydli, List, and Suskind 2017; List and Metcalfe 2014).

proposals of diversity recruitment policies try to address identity concerns by featuring male workers in advertisements for pink-collar jobs, such as nursing or teaching (*Economist* 2018).[3]

The first experimental treatment mimics these policy proposals by including photographs of real workers in the recruitment messages and randomizing whether the portrayed worker is a man or a woman (labeled "Male Photo" and "Female Photo" treatments, respectively). Surprisingly, I find that the photographs do not affect men's applications. This casts doubts on the effectiveness of varying workers' gender composition in hiring campaigns to attract more men into female-dominated jobs. While organizations may want to showcase diversity for other reasons (e.g., reputation), these strategies are unlikely to affect men's selection.

Alternatively, men may not know whether they can do a good job or if their talent will be valued in female-dominated jobs. As minorities in a particular occupation, they may have noisier expectations due to receiving fewer signals about the work environment and how people perform. The caring nature of female-dominated jobs may further deter men from getting informed or engaging in them, if they fear not being successful and choosing a career that is not ambitious (Goldin 2006; Simpson 2009).[4] Survey data that I collected indeed show that men have high uncertainty about their potential performance and tend to underestimate the rewards that talented workers can realize in female-dominated occupations.

To address this informational disadvantage, the second treatment involves providing information in the recruitment messages on the share of workers who received high performance evaluations in their service delivery. In a randomized way, one-half of the sample are informed that 66 percent of workers achieved the highest evaluation scores, and the other half that 89 percent of workers achieved the highest evaluation scores in interacting with their customers in a previous year. I use the partner organization's HR records to communicate truthful but partial information (Dal Bó, Dal Bó, and Eyster 2017). I label these treatments "Lower Share" and "Higher Share," respectively.

I find that providing information on the share of high-performing workers affects men's applications, the applicant pool quality, and the downstream performance that new male hires have on the job. At the application stage, men are more likely to apply in the "Lower Share" group. Applications increase by 14 percent with respect to receiving the "Higher Share" ($p = 0.03$) and by 11 percent with respect to a pure control group that received messages with no experimental manipulation ($p = 0.15$). Qualitatively, the "Lower Share" impact is stronger among men with high job-specific talent and better outside options. Finding out that the past rate of high performers was relatively low (at 66 percent) seems to encourage good male candidates to apply.

When examining the pool of applicants (i.e., candidates who self-select into applying), men in the "Lower Share" group have indeed better observable characteristics and receive more job offers than men in the "Higher Share" (by

[3] For instance, portraying more male nurses is one of the pillars of the biggest recruitment drive in the history of the UK National Health System (NHS), see McGonagle (2019).

[4] In psychology and sociology, the need to feel effective is a key factor in career choices, especially those incongruent with one's own gender (Dweck and Elliot 2005; Elliot, McGregor, and Thrash 2002; Skorikov and Vondracek 2011; Hirschi 2012).

5 percentage points, or 45 percent of the pure control mean). This gap in offer rates arises from both a decrease of applicants' quality in the "Higher Share" group and an increase in quality at the very top of the ability distribution in the "Lower Share." This suggests that a workplace where everyone performs well attracts more unsuitable applicants, while an indication that there is room for improving past performance unlocks applications from some very talented marginal applicants.

Even though the sample size is small and the following result should be taken with caution, the "Lower Share" seems to ultimately enable the employer to hire and retain better male workers. Over the first two years on the job, men in the "Lower Share" group perform better (by 9 percent of the pure control mean) and have more positive attitudes about the job than workers in all other experimental groups. This evidence indicates that the "Lower Share" can improve gender diversity with an additional benefit on the quality of incoming workers. Through its impact on attitudes, the treatment may also help workers sustain their motivation on the job (Grant 2008).

Furthermore, I find that disclosing the "Lower Share" information remains desirable for the employer after accounting for women's reaction to it. On average, the information treatment has no impact on women's applications, leading me to reject the hypothesis that the two genders react similarly to the "Lower Share" ($p = 0.02$). However, the "Male Photo" discourages women's applications as well as retention once hired. These negative effects are driven by low-ability women and are concentrated in the group that received both the "Male Photo" and the "Lower Share" treatments. Therefore, the combination of the two treatments improves the quality of women who stay in the job, but it also raises turnover costs. A survey conducted with the personnel of the partner organization revealed that most of them deemed the increase in performance to be high enough to justify the higher turnover. Thus, the effect of the informational policy that attracts more men is positive for the employer, even net of female turnover costs.

To investigate the main interpretations of the photograph and information manipulations, I conducted survey experiments with out-of-trial samples.[5] While each treatment may have multiple interpretations, the surveys shed light on the leading mechanisms and help rule out unrelated explanations. For both men and women, the photograph manipulation is efficacious in changing some key gender-related beliefs about the job, such as perceived male shares and the employer's preferences for men. However, photographs do not affect men's opinion on the desirability of social work for their gender or their scores in a "women–social care" implicit association test (IAT), while they influence women's opinions on both dimensions. The fact that the association between women and social work is sticky for men—but not for women—might explain why the use of male photographs has a null effect on men's entry into the job, whereas it has a negative impact on women's applications.

Regarding the information manipulation, I find that the two treatments create a wedge in beliefs on the share of successful workers between groups, as intended by design (a gap of 8 percentage points, $p = 0.00$). However, this does not translate

[5] Respondents were applicants for the partner organization's job (one year after the main experiment) as well as an online population of UK students and workers matched on observables with the field sample.

into a straightforward updating of men's expectations of success. I rule out that the manipulation affects the perceived difficulty of job tasks or the probability of getting the job. Perhaps less intuitively, I find that the "Lower Share" increases the expected ranking for the job for high-ability relative to low-ability respondents. This is consistent with the provided information either signaling a more competitive workplace—where good performance is assessed and rewarded—or indicating there is room for talent to improve past effectiveness in service delivery.

The positive impact of the "Lower Share" on men's applications and the interpretation emerging from the surveys suggest that rewarding talented workers' contribution—relative to the untalented—is attractive for skilled male candidates. This result contradicts the intuition that informing minorities of high past success should increase their expected performance in jobs that are uncommon for their group (Porter and Serra 2020; Breda et al. 2023; Del Carpio and Guadalupe 2022). Previous research, however, has focused on male-dominated occupations, which typically have steep incentives. In sectors where people expect flat incentives, such as social work, high success may reinforce perceptions of low rewards for talent and encourage mostly low-ability people to apply, as observed in the "Higher Share" treatment.

Compared to similar light-touch interventions, the magnitude of the "Lower Share" effect on applications is also substantial, which is noteworthy given that the treatment is cost free for the employer. One plausible reason is that men have high uncertainty about the job and thus react strongly to new information. This is supported by evidence that the treatment has the strongest impact on applications by men with limited exposure to social work or female-dominated jobs (Jensen 2010; Charles, Guryan, and Pan 2022; Baranov, De Haas, and Grosjean 2023).

Taken together, my results suggest that breaking informational barriers to men's entry into female-dominated jobs might increase gender diversity and improve workforce quality, yielding an optimistic message for policy. The stigma associated with men working in female-dominated occupations and their perceptions of potential success have been central in the debate about converting unemployed men into female-dominated jobs (Miller 2017). The policy implications for these barriers differ. Modifying deeply ingrained gender roles in certain occupations may be challenging, yet organizations can use strategies like quotas to drive compositional changes in their workforce. Alternatively, uncertain or incorrect expectations about female-dominated roles can be more cost-effectively tackled through information provision and changes in incentives, for example, using low-cost organizational practices that recognize good performance. While increasing diversity, a focus on ways to reward performance could also help employers overcome the challenges of finding and retaining talent (Delfgaauw and Dur 2010; Finan, Olken, and Pande 2017).

This paper contributes to the growing empirical work on the impact of stereotypes and social identity on career choices (Hoff and Pandey 2006; Bordalo et al. 2016; Glover, Pallais, and Pariente 2017; Alan, Ertac, and Mumcu 2018; Carlana 2019; Oh 2023; Coffman, Collis, and Kulkarni 2023a; Del Carpio and Guadalupe 2022; Palffy, Lehnert, and Backes-Gellner 2023). Previous work has analyzed the overall impact of identity on choices. However, group belonging (e.g., gender) and group-specific expected success jointly determine the identity payoffs that a person enjoys in a certain task or occupation. We have little knowledge on how these two economic fundamentals compare in explaining behavioral differences between groups. This paper takes a step in this direction by proposing a novel design that disentangles these factors.

Relatedly, while studies about gender differences across male- or female-typical domains abound in the lab (for a review, see Azmat and Petrongolo 2014), tasks in the latter tend to be stylized. I study the determinants of men's choices in a real female-typical field context. Not only is this setting relevant from a policy perspective (Katz 2014), but it is also high stakes, as participants are making a real-life choice with far-reaching implications for their careers.

The paper also contributes to the personnel economics literature on the effects of advertised job amenities or requirements on applications and hiring (Dal Bó, Finan, and Rossi 2013; Marinescu and Wolthoff 2020; Ashraf et al. 2020; Deserranno 2019; Abebe, Caria, and Ortiz-Ospina 2021; Flory et al. 2021; Coffman, Collis, and Kulkarni 2023b; Abraham, Hallermeier, and Stein 2022; Kuhn and Shen 2023; Card, Colella, and Lalive forthcoming). As in Flory et al. (2021) and Abraham, Hallermeier, and Stein (2022), this paper focuses on recruitment strategies aimed at increasing organizational diversity. I expand on this work by following hires on the job, allowing me to quantify the net benefit of increasing diversity for the employer and to uncover the existence of long-term effects of diversity recruitment policies (Azmat and Boring 2020). Notably, the paper also sheds light on the importance of reducing uncertainty in men's expected success, complementing the findings of Coffman, Collis, and Kulkarni (2023b), who find that reducing ambiguity related to job fit encourages women into male-dominated fields.

I. Institutional Context

Social Work: Social work is a good setting for my research questions for several reasons. Historically, women have represented more than 70 percent of social workers in the United States and United Kingdom (online Appendix Figure A.1, panel A). As with other female-dominated service occupations, the demand for social workers is growing (online Appendix Figure A.1, panel B). Projections indicate that the field will grow at double the average growth rate of all other US occupations in the next decade (US Bureau of Labor Statistics 2019), with peaks in regions with high rates of male joblessness (online Appendix Figure A.1, panel C). However, this growth trajectory is challenged by looming staff shortages (Lin, Lin, and Zhang 2016), further complicated by difficulties in retaining existing social workers. In the United Kingdom, nearly half of social workers plan to leave their jobs within two years, primarily due to excessive workloads (Ravalier 2019).

In addition, a consistently high share of female workers may create uncertainty or pessimism about men's performance in social work. This is supported by survey evidence that I collected with UK students and applicants for the partner organization's job. Online Appendix Figure A.2 shows that respondents expect men to perform significantly worse than women in social work, although their confidence in this expectation is low. Moreover, men are even more pessimistic and less confident than women. In a separate sample of UK students and workers, similar patterns emerge regarding beliefs about the returns to talent in female-dominated roles. Both genders underestimate the relative rewards that a talented individual can achieve compared to an untalented one. Once again, men are more pessimistic about these returns for their own gender, suggesting a potential lack of information regarding their chances of success in social work.

Partnership: During 2017, I collaborated with one of the main UK recruiters of public sector social workers. The organization offers a two-year on-the-job training position targeted at either final-year students from all disciplines or current workers from any industry. Previous experience in social work is not required, allowing me to study entry into this occupation broadly, not only in a sample with job-specific training. Moreover, informational and psychological frictions may be especially relevant for the portion of the sample with limited prior exposure to social work.

Every year, new hires are assigned to teams located in local authorities across England and earn a stipend that is comparable to the average UK annual entry salary in social services, primary school teaching, or nursing. The job involves office tasks (e.g., case writing) and meetings with families in need and other stakeholders, such as lawyers or doctors. After the program, around 80 percent of workers stay in social work, while most of the others move to policymaking roles.

The program is part of the movement toward the professionalization of the public sector. As such, the organization's key challenges are related to the attraction, retention, and diversity of talent. Attracting gender diversity without sacrificing talent or retention was also the goal of the randomized controlled trial (RCT) we designed. Figure 1 illustrates the timeline of the organization's 2017 nationwide recruitment. The experiment took place during the application period, which ran from September to November. The hiring process included several stages, conducted either online or at the organization's headquarters, with candidate screening occurring at each step. For an applicant, the entire process—from application to job offer—typically lasted about ten weeks. Those who were hired and accepted the job started working in local authorities in July 2018. I followed these workers throughout the program until July 2020.

II. Experimental Design

Sampling: Experimental participants are individuals who are interested in applying for the job offered by the partner organization. To express their interest, they complete a brief registration form on the organization's website, including eligibility and demographic questions, which typically takes three to five minutes to complete. If eligible, respondents receive an email inviting them to apply.[6] The need to register implies that the experimental sample is selected on a minimum level of interest for the job. However, application rates after registration are around 50 percent for men and 60 percent for women, indicating there is substantial room for changing selection into the job within this sample. Furthermore, from a policy perspective, this group may be precisely the relevant target to try new recruitment strategies. Section IIIC further describes the sample.

[6] The email contains a person's candidate number, which is needed to enter the application portal, and information about the hiring process. Respondents who do not meet the eligibility requirements receive a standard rejection email. Eligible applicants should have a bachelor's degree with a certain minimum average grade and have obtained at least a C in math and English pre-university qualifications. Social work students are not eligible.

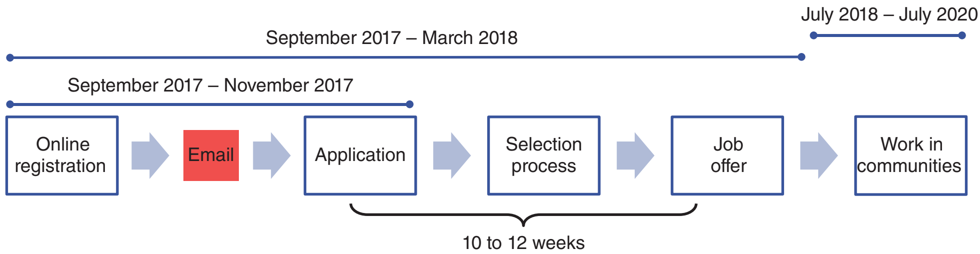

Figure 1. Recruitment Timeline

Notes: The figure shows the partner organization's recruitment timeline. Applications were open from September to November 2017. Candidates were randomized in one of the different invitation-to-apply emails right after their online registration. The organization completed the recruitment process in March 2018. For a successful applicant, it took around ten weeks between application and job offer. If a person was hired and accepted the job, actual work in local communities started in July 2018. Performance data are collected for the entire duration of the two-year program between July 2018 and July 2020.

Experimental Design: I introduce exogenous variation in the content of the invitation-to-apply email along two dimensions: photographs and information. The two experimental conditions were cross-randomized in a fully nested design, leading to a total of four treatment emails (see Figure 2 for an example of a treatment email). Participants could also be randomly assigned to receive a fifth email containing no manipulation (pure control), which represents business as usual for the organization. Randomization was done at the individual level, with stratification by gender (man/woman), ethnicity (white/nonwhite), and whether a person registered during the week before the official opening date.

The experiment was double-blinded: participants were not aware that the invitation-to-apply email was part of a research study, and recruiters were not aware of candidates' treatment assignment. This design limits biases that arise from candidates' knowledge of being in a research study and prevents recruiters' assessment from being influenced by the candidates' treatment. I discuss each of the experimental manipulations below.

Photograph Treatment.—The invitation-to-apply email included a photograph of an actual worker, randomized to be either a man or a woman. The design draws on audit and priming studies (Bertrand and Mullainathan 2004; Benjamin, Choi, and Strickland 2010), in that seeing a male photograph should weaken identity concerns among men by changing gender-related perceptions about the job (e.g., perceived male shares, salience of the gender stereotype, employer's preferences).

I assigned different photographs to white and nonwhite individuals and matched the ethnicity of the photographed workers with that of each candidate (while randomizing gender). Without this matching, it would be difficult to determine whether the effects are attributable to gender or ethnicity. For instance, if white male candidates are more likely to apply after seeing an email featuring a white man than after one featuring a nonwhite woman, I would not be able to know whether the effect is due to the gender or ethnicity match. Moreover, showing photographs of white individuals at the start of the hiring process could trigger stereotype threat among nonwhite candidates (Steele and Aronson 1995), who represent almost 30 percent of the sample.

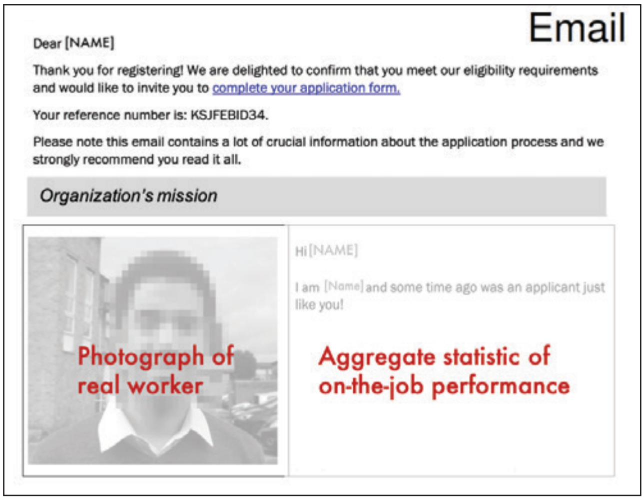

Figure 2. Intervention Email Template

Notes: The figure shows a stylized example of one of the email templates used in the intervention. The intervention box containing both the photograph and information manipulations was located in the top quarter of the email and could be visualized in the email preview on any smartphone or tablet. To further attract attention to it, the box was positioned below the candidate number, which was needed to access the application portal, and the text to the right of the picture addressed the candidates by name. The email footer was signed by the Director of Selection and showed the organization's logo and a disclaimer of confidentiality. The pure control email did not show any intervention box.

The design of this manipulation addresses other potential confounders. To attract the candidate's attention to the photograph, I added a short text where the photographed worker addresses the candidate by name and recalls that they were also once an applicant. Drawing on studies on role models (Marx and Ko 2012) and information retrieval (Schwarz et al. 1991), this message should facilitate the candidate's relatability to the portrayed person and the gender group they belong to. In addition, the photographed people were real workers who were not featured in other advertisements or media contents of the organization. This limits the impact of unobserved heterogeneity in candidates' exposure to the organization's recruitment materials. All photographs show the same background and are of the same size to limit visual differences.

Given constraints in the availability of workers of different genders and ethnicities, I use one photograph in each treatment-ethnicity cell, increasing the risk of a systematic correlation between portrayed workers' characteristics and their gender. To mitigate concerns about this potential bias, I asked three survey samples (described in online Appendix C) to rate the photographs on several dimensions. The portrayed workers are deemed similar to one another in work satisfaction and trust, but the white man is perceived as less friendly or happy than the white woman (online Appendix Table A.1). However, these differences are unlikely to drive the main results.[7]

[7] Online Appendix Table A.2 shows that the photographs' impact on applications is the same for white and nonwhite candidates. Thus, differences in friendliness cannot explain the effects of photographs, as the male and female nonwhite workers are rated equally friendly and happy.

Information Treatment.—This manipulation communicated the outcome of a selected past cohort of workers, which had either a higher or lower share of workers who achieved high evaluations. The exact wording was the following: "Did you know that in one of the past cohorts X percent of participants got commendable or excellent feedback to their interaction with families?" where X was equal to 66 or 89 in the two experimental treatments. Commendable or excellent are the highest grades that people can achieve in their performance assessments in the job, which are given by experts in social work (external to the organization). The experimental information refers to the evaluation of workers when interacting with their customers (i.e., families), and thus, it relates to their performance in service delivery.

Both statistics were taken from the partner organization's HR records. I calculated the percentage of workers in each cohort (i.e., all workers hired in the same year) who received evaluations exceeding 60 out of 100, which was the threshold recommended by the organization for remarkable performance. I then selected the two cohorts with the highest and lowest percentages, creating the largest gap between treatments. This enabled the communication of truthful but partial information, which on average should create a wedge in the beliefs of success between experimental groups independently of priors (Dal Bó, Dal Bó, and Eyster 2017; Settele 2022).[8]

I reported information about on-the-job success in frontline interactions with clients for several reasons, primarily to induce variation in people's beliefs concerning their potential success in one of the most important tasks on the job. The quality of clients' interactions is a crucial objective in the organization's mission, and candidates care about it when applying (Besley and Ghatak 2005).[9] Additionally, performance in service delivery is rarely disseminated in the industry, a fact that increases the likelihood that the provided information will affect a candidate's beliefs. Finally, service delivery outcomes also depend on clients' reactions. A low score can signal clients' hostility and/or discrimination, which can disproportionately affect men's and other minorities' assessment of their potential job success.

III. Empirical Strategy, Sample, and Balance

A. *Empirical Strategy*

My main empirical strategy relies on the independent random assignment of the photograph and information manipulations and compares the impact of one type of photograph (or type of information) against the other in affecting men's entry and selection into social work. I estimate the following regression, separately for men and women:

$$(1) \qquad y_i = c + \beta_1 MalePhoto_i + \beta_2 LowerShare_i + \mathbf{X}_i'\lambda + \epsilon_i,$$

[8] Using data from out-of-trial surveys (described in Section V), online Appendix Figure A.11, panel B shows that respondents understood the experimental information: both men and women think that there is a higher share of successful workers in the job when told that 89 percent rather than 66 percent of participants got the highest feedback (a gap of 6 and 9 percentage points, respectively, $p < 0.01$ for both genders). Priors are statistically indistinguishable from posteriors in the "Higher Share" treatment for both genders.

[9] To emphasize the connection with the organization's mission, the intervention box was placed just below a summary of its purpose statement, which is focused on improving outcomes for disadvantaged communities.

where $MalePhoto_i$ is equal to one if potential applicant i is assigned to receive a male photograph (zero for the female photograph) and $LowerShare_i$ is an indicator for receiving information of a lower share of high performers (zero for the higher share). $\mathbf{X}_i$ contains controls chosen with Post-Double Selection Lasso (Belloni, Chernozhukov, and Hansen 2014) among baseline variables. Randomization was done at the individual level, so I use Eicker-Huber-White robust standard errors. For robustness, I use randomization inference (Young 2018) and adjust standard errors for multiple-hypothesis testing (MHT) within each manipulation and gender (Benjamini, Krieger, and Yekutieli 2006; List, Shaikh, and Xu 2019).

β_1 tests the null hypothesis of no effect of photographs on outcome y_i, conditional on also getting information on workers' evaluations. β_2 measures how outcome y_i changes in the "Lower Share" vis-à-vis "Higher Share" treatment, conditional on also getting a worker's photograph. β_1 and β_2 identify the causal effect of photographs and information on y_i, respectively, under the assumption of no interaction between the two manipulations. This assumption is empirically valid for men (online Appendix Table A.6), although the statistical power to check this is limited.

While an email without any manipulation is a useful counterfactual for recruiters, interpreting the comparison with the pure control is not straightforward because each treatment email jointly changes information and photographs. Moreover, the treatment-control comparison is less precise than a comparison between treatments because it does not account for possible side effects of giving information (e.g., salience on certain job features) and depends on priors in the control group, which are possibly noisy and correlated with individual unobservables (Settele 2022). Nonetheless, I show results for the pure control in all the figures and report specifications comparing each treatment with the pure control in the online Appendix. To increase power, these regressions also use double machine learning (Chernozhukov et al. 2018) for covariate selection.

B. *Main Outcome Variables*

I consider two primary sets of outcome variables y_i: application outcomes (entry and quality) and on-the-job outcomes (performance and retention).[10]

Application Outcomes: My outcome variable for entry equals one if a candidate applies and does not voluntarily withdraw at any stage of the hiring process. To assess whether the treatment attracts better- or worse-skilled applicants, I consider whether i receives a job offer and whether they accept the offer (both coded as zero if i did not apply).

On-the-Job Outcomes: New hires are continuously evaluated throughout the two-year program. I measure performance on the job for worker i as the weighted average test score, where weights are given by the credits that the organization assigns to each test. Performing well on these tests is important to be able to qualify

[10] My pre-registration lists the sets of primary outcomes and the treatments, which are the ingredients needed in the main specification (which was not pre-registered). The same registration includes the outcomes and treatments of the experiment described in online Appendix D.

as a social worker.[11] To measure turnover, I define an indicator for quitting the program before the end. I further explore attitudes toward the job (e.g., job satisfaction) through employee surveys collected by the organization every semester.

Identification Assumptions.—To interpret differences in outcomes as the causal effect of the treatment on applicants' self-selection, the identification assumption is that the individual probability of being successful at each stage of the hiring process is independent of treatment assignment. This assumption could be violated if the treatments influence candidates' effort or the employer's screening criteria. The double-blind design prevents the employer from being affected by the treatment assignment. Nonetheless, Section VC discusses both concerns.

C. *Sample and Balance*

The experimental sample comprises 5,417 candidates, with 1,013 men. Among them, 807 men and 3,513 women are assigned to one of the treatments. Table 1 presents summary statistics by gender and balance checks for the full sample. The average age of candidates is 27, and 3 out of 10 are ethnically nonwhite. While 32 percent of the candidates studied in a top-tier UK university, the share of people from lower socioeconomic backgrounds is also substantial. Within the sample, 19 percent of candidates have parents in unskilled occupations, 27 percent received school financial aid, and 2 percent were looked after by social workers during childhood (compared to 0.12 percent in the United Kingdom).

Male and female candidates have similar socioeconomic backgrounds and exposure to the organization but differ in terms of demographics, education, and employment. Men are older and therefore more likely to have graduated or to work full time (FTE). The same share of men and women attended a top-tier UK university or got a first grade, but men are more likely to have studied scientific subjects and, if working, to be in corporate or scientific jobs.

Treatment assignment is balanced on observables. This is shown in columns 5 and 6 of Table 1, which report the F-statistics and the related p-values of a regression for each of the row variables on the set of four treatment indicators. The last column reports the minimum p-value of pairwise t-tests for the difference in means between each pair of treatments along the variables reported. Only 3 p-values are lower than 0.05 out of 260 tests. I also fail to reject the null hypothesis of zero effect of all the variables reported in the table in a joint test of orthogonality on assignment to any treatment group ($F(26, 4{,}959) = 0.7$).

Online Appendix Table A.3 compares the experimental sample with UK students (panel A) and workers (panel B). Panel A shows that participants are less likely to study STEM or business and more likely to be trained in languages, the humanities, and social studies compared to the average UK student. Panel B shows that participants are more likely to be of nonwhite ethnicity and to work in the public

[11] Assessments include theory (e.g., case studies, essays) and practice tests, and scores are given on a scale from 0 to 100. Theory tests are evaluated anonymously by experts in social work. Anonymity is not possible in practice tests, but evaluators were not aware of candidates' treatment or that the experiment occurred. Practice assessments include direct observations of workers' interaction with customers and a first-month review done by teachers at the end of a mandatory classroom training to evaluate each worker's potential for the job.

Table 1—Balance Checks and Summary Statistics

	Men ($N = 1{,}013$)		Women ($N = 4{,}404$)		Joint		Pairwise
Variables	Mean (1)	SD (2)	Mean (3)	SD (4)	F-stat (5)	p-val (6)	min p-val (7)
Male	1.00	0.00	0.00	0.00	0.04	1.00	0.72
Nonwhite	0.28	0.45	0.27	0.45	0.08	0.99	0.60
Age	28.68	9.17	26.35	7.94	0.29	0.88	0.42
Married ($N = 995, 4{,}331$)	0.19	0.40	0.12	0.33	0.18	0.95	0.47
Caring duties	0.16	0.36	0.16	0.37	0.96	0.43	0.11
LGBT ($N = 959, 4{,}131$)	0.13	0.34	0.07	0.26	0.36	0.84	0.33
Non-British	0.18	0.38	0.13	0.34	0.47	0.76	0.21
Religious	0.44	0.50	0.46	0.50	0.05	1.00	0.68
Past application	0.07	0.26	0.06	0.24	0.08	0.99	0.61
Pre-submission call	0.11	0.32	0.08	0.28	0.48	0.75	0.27
Early registration	0.04	0.20	0.05	0.21	0.31	0.87	0.39
Registration before November	0.53	0.50	0.57	0.50	0.02	1.00	0.83
Any event	0.00	0.05	0.01	0.11	0.13	0.97	0.52
Top UK university	0.33	0.47	0.32	0.47	0.21	0.94	0.38
First grade	0.20	0.40	0.18	0.39	0.70	0.59	0.13
Graduate	0.46	0.50	0.35	0.48	0.35	0.84	0.29
Grade is expected	0.25	0.43	0.38	0.49	0.04	1.00	0.70
Full-time employment (FTE)	0.49	0.50	0.42	0.49	0.25	0.91	0.41
in: health care	0.15	0.36	0.16	0.37	0.85	0.49	0.11
in: public sector	0.45	0.50	0.55	0.50	1.07	0.37	0.05
in: corporate/business	0.31	0.46	0.21	0.41	1.17	0.32	0.04
Scientific subject	0.09	0.28	0.05	0.21	0.24	0.92	0.40
Economic school support	0.27	0.44	0.27	0.45	0.62	0.65	0.15
Low socioeconomic status	0.60	0.49	0.62	0.49	0.85	0.49	0.08
Young carer ($N = 999, 4{,}339$)	0.04	0.20	0.04	0.20	1.68	0.15	0.02
Care leaver ($N = 1{,}006, 4{,}369$)	0.03	0.17	0.02	0.15	0.46	0.76	0.26

Notes: The table shows summary statistics and balance checks of baseline variables (from the registration form). Columns 5 and 6 report the F-statistic and p-value from a joint test of the significance of the four treatment dummies in explaining each row variable, with robust standard errors. Column 7 reports the minimum p-value of t-tests between pairs of treatment groups. "Top UK" universities belong to the Russell Group. "First grade" and "Grade is expected" are dummies for achieving the highest university grade and not having official confirmation of the university grade. "Past application" and "Pre-submission call" equal one for candidates who previously applied for the job and were called by a recruitment officer before applying. "Early registration" and "Any event" are dummies for having access to an early opening of the application and for attending one of the organization's events. "Economic school support" equals one if the candidate received government support in school. "Low socioeconomic status" equals one if the highest earner in one's household had a low-skilled job or for parents with no degree. "Care leaver" and "Young carer" are dummies equal to one if the person was cared for by a social worker before the age of 14, and if they ever cared for a relative. In parentheses are the number of male and female respondents for a given variable, if different from the total.

sector and less likely to be married or to have completed their studies. This evidence suggests that participants in the experiment may have a preference for working in female-dominated jobs and be more informed about them. Consequently, the estimated effects for both photographs and information might be downward biased relative to the effects expected in a broader population, as the intervention may not be surprising enough in my sample. However, it is also possible that both treatments would have weaker impact in the general population, for instance, if reduced interest in social work decreases attention toward recruitment messages.[12]

[12] To rule out a differential bias in the estimates between treatment groups, I assume that the sample is equally selected on attitudes toward gender-related factors and information throughout the analysis.

IV. Experimental Results

This section presents the main results of the paper. I first look at men's applications, hiring, job performance, and retention. I then discuss treatment effects on women's outcomes, which allows me to compute the net impact of the intervention for the employer. I conclude by exploring heterogeneous treatment effects in application behavior.

A. *Men*

Application Outcomes.—Table 2, panel A summarizes men's outcomes in the hiring process.[13] The first result indicates that the photograph manipulation does not affect men's applications (column 1). Although being in the "Male Photo" group reduces men's applications by 2.3 percentage points compared to the "Female Photo," this effect is indistinguishable from 0 ($p = 0.50$). The second result reveals that men react to the information manipulation. As shown in the second row of column 1, men's applications increase in the "Lower Share" group by 7.5 percentage points ($p = 0.03$). This increase represents 14.4 percent of the "Higher Share" application rate (52 percent) and 14.2 percent of the pure control mean (53 percent).

To disentangle different channels, online Appendix Table A.6 compares each of the four treatment emails with the pure control. Neither the male nor the female photographs affect men's applications compared to the business-as-usual email. The effect on men's applications of the "Male Photo"—averaged across information treatments—is 0.01 ($p = 0.80$) and of the "Female Photo" is 0.03 ($p = 0.46$) compared to the pure control. Men's application rates in the "Higher Share" group are also similar to the pure control, with an average effect across photographs of -0.02 ($p = 0.67$). In contrast, compared to no manipulation at all, men's applications increase by 5 to 7 percentage points in the "Lower Share" group when combined with a male or female photograph, which represent 9 percent to 13 percent of the pure control mean (randomization inference $p = 0.29$ and $p = 0.14$, respectively). Thus, men seem to be encouraged to apply in the "Lower Share" group rather than be discouraged in the "Higher Share" one. This finding aligns with the observation that, on average, people's prior beliefs about workers' performance are close to their posteriors when receiving the 89 percent statistics (see online Appendix Figure A.11, panel B), making the "Lower Share" information unexpected.

The last two columns of Table 2 show hiring outcomes. Looking beyond applications is crucial because the treatments may influence the quality of men who self-select into applying. Despite no impact on applications, columns 2 and 3 show a positive effect of the male photograph on men's job offers (by 3.3 percentage points, $p = 0.09$) and acceptance (by 2.8 percentage points, $p = 0.08$) compared to the female one. However, this gap is driven by lower offer and acceptance rates

[13] Online Appendix Table A.4 presents estimates with only strata controls, and online Appendix Table A.5 employs various outcome definitions (e.g., applications and withdrawals separately, conditional job offers). Results align closely with those in Table 2.

Table 2—Application Outcomes

	Application outcomes		
	Applied and no dropout (1)	Received job offer (2)	Accepted job (3)
Panel A. Men			
Male Photo	−0.023	0.033	0.028
	(0.035)	(0.020)	(0.016)
	[0.50]	[0.09]	[0.08]
	{0.53}	{0.22}	{0.20}
Lower Share	0.075	0.049	0.027
	(0.035)	(0.020)	(0.016)
	[0.03]	[0.01]	[0.11]
	{0.07}	{0.04}	{0.09}
Photo = info p	0.05	0.57	0.94
Mean DV	0.52	0.05	0.03
Mean DV in PC	0.53	0.11	0.09
Observations	807	807	807
Panel B. Women			
Male Photo	−0.051	−0.001	0.011
	(0.016)	(0.009)	(0.008)
	[0.00]	[0.94]	[0.14]
	{0.01}	{0.93}	{0.22}
Lower Share	−0.014	−0.001	−0.000
	(0.016)	(0.009)	(0.008)
	[0.41]	[0.90]	[0.97]
	{0.69}	{0.97}	{0.97}
Photo = info p	0.11	0.96	0.30
Mean DV	0.60	0.09	0.05
Mean DV in PC	0.59	0.09	0.06
Observations	3,513	3,513	3,513
Gender differences			
Photo: men = women p	0.46	0.12	0.34
Info: men = women p	0.02	0.02	0.13

Notes: The table shows OLS estimates with controls chosen using PDS Lasso. The dependent variables in columns 1 to 3 are indicators for applying and never dropping out of the process, receiving a job offer, and accepting the job offer. Square brackets contain the p-values from randomization inference with 1,000 repetitions, and curly brackets report FWER-corrected p-values (List, Shaikh, and Xu 2019). The row "Photo = info p" reports the p-values from the test of equality of coefficients "Male Photo" = "Lower Share," within each gender sample. "Mean DV" and "Mean DV in PC" are the mean of the outcome variable in the omitted category and the pure control group, respectively. The rows under "Gender differences" report p-values from a Wald test of coefficient equality between the men's and women's regressions, separately for the "Male Photo" and "Lower Share" coefficients.

in the "Female Photo" compared to the pure control, while the "Male Photo" has no impact on either variable (online Appendix Table A.6). The "Female Photo" thus attracts lower-quality male applicants, who are less likely to get the job and are also less motivated to accept it.

The gap in applications created by the information manipulation translates into differences in hiring. Men in the "Lower Share" group get more job offers and are slightly more likely to accept the job compared to men in the "Higher Share" group. Offer rates increase from 5 percent in the "Higher Share" to 10 percent in the "Lower Share" ($p = 0.01$), and acceptance rates increase from 3 percent to 5.7 percent

($p = 0.11$). The difference in offer rates may come from an improvement in the quality of applicants in the "Lower Share" or a decrease in the "Higher Share" group. The following exercises provide evidence of both channels being at play.

First, applicants in the "Lower Share" treatment score higher in an index of "desirable skills," which are observable correlates of receiving a job offer (online Appendix Figure A.4, panel A, K-S test $p = 0.01$). Second, quality at the top of the applicants' distribution also differs across information treatments, a fact consistent with a change in the self-selection of the marginal applicant. In addition to the index of desirable skills defined above, I measure quality as the predicted on-the-job performance based on observables, the average test score in the hiring process, and an index of the three measures.[14] To identify top applicants, I follow the employer's screening rules, which were blind to treatment status and created a unique ranking of candidates based on their performance in the hiring tests, independent of gender. The top quintile identifies applicants who would have been hired by the employer, given slot availability. Online Appendix Figure A.5, panel A shows that within this quintile, the highest quality is achieved in the "Lower Share" group. Higher quality, in turn, is correlated with more job offers at the top: 54.4 percent in the "Lower Share" compared to 50 percent and 41 percent in "Higher Share" or pure control (differences are not statistically significant, $observations = 130$).

I also find evidence of a decrease in applicants' quality in the "Higher Share" group. While 70 percent of job offers are made to applicants in the top quintile of the overall ranking, marginal offers are made in the second and third quintiles. Online Appendix Figure A.6 shows that when considering these three quintiles together, applicants' quality in the "Higher Share" group is lower than both the pure control and "Lower Share" groups. This means there is a mass of lower-talent candidates who apply when receiving information that almost everyone performs well in the job and who push down the quality of potential hires in the "Higher Share" group. Consequently, the overall rate of job offers in the "Higher Share" treatment is significantly lower than in the pure control group by 5 percentage points (column 2 of online Appendix Table A.6, $p = 0.05$). Not only do men in the "Higher Share" group receive fewer job offers than men in the pure control, but they are also less likely to accept them by 5 percentage points (column 3 of online Appendix Table A.6, $p = 0.02$).[15]

To summarize, relative to the "Higher Share" information, the "Lower Share" attracts more and better male applicants, who are consequently more likely to be hired. Even if the overall offer rate in the "Lower Share" group is the same as in the pure control (online Appendix Table A.6), the share of top candidates who get offered a position is qualitatively larger, and their quality higher, than in the group who got the business-as-usual email.

[14] Online Appendix B.3 describes the construction of predicted on-the-job performance. All the indexes are standardized to have mean zero and unitary standard deviation in the pooled sample of men and women (Kling, Liebman, and Katz 2007).

[15] Theoretically, changes in applicants' quality with no change in the size of the pool are possible. A simultaneous decrease in applications by top candidates and an increase by bottom ones could lead to an overall decrease in quality, even without changes in numbers. This is consistent with the empirical evidence for the "Higher Share" as well as the "Female Photo" in comparison with the pure control.

The information treatment may also change the types of people who apply for the job beyond what is captured by quality metrics. Online Appendix Figure A.7 shows the relationship between application and baseline demographics, education, or employment in each information treatment. In the pure control, none of the variables listed in the figure are selected when performing a Lasso of application on observables, indicating they are not good predictors for applying. Nevertheless, the coefficient's sign on a few observables is reversed in the "Lower Share" group compared to the pure control. For instance, health care experience positively predicts application in the pure control but negatively predicts it in the "Lower Share" group. Therefore, by increasing men's entry, the "Lower Share" treatment appears to also weaken the standard correlates of application behavior and increase applicants' diversity of background and experience.

On-the-Job Outcomes.—Table 3, panel A shows on-the-job outcomes for men. The sample for this analysis comprises the universe of new male hires, but it is small—43 out of 67 job offerees in the treatment groups—and thus, the results should be taken with caution.[16] Given the limited sample, I do not use PDS Lasso but only control for strata and working region fixed effects in this analysis.

Columns 1 and 2 of Table 3 show that male hires in the "Lower Share" group perform better on the job. The difference in test scores between the two information treatments is 4.28 percentage points over 2 years (out of a pure control mean of 53, $p = 0.21$). The gap in male performance between the "Lower Share" and the "Higher Share" is 9.5 percent of the pure control mean during the first semester ($p = 0.03$) and 9 percent in the last year, but it becomes insignificant ($p = 0.51$). Assuming no spillovers on inframarginal workers, the performance of hires induced to apply by the "Lower Share" is 23 percent higher than those in the alternative information treatment in the first semester.[17]

Men attracted to apply by the "Lower Share" treatment perform better along the entire distribution of on-the-job performance. Figure 3, panel A shows that the distribution of average test scores is shifted to the right for men who received the "Lower Share" information vis-à-vis the "Higher Share" (K-S test $p = 0.06$). Similar distributional differences are observed with respect to the pure control, but they are estimated with noise (K-S test $p = 0.21$).

Quality metrics (e.g., observable skills or hiring test scores) account for little of the gap in performance between treatment groups (online Appendix Figure A.8, panel A). The residual difference may be explained by a wider range of skills that are not observable to the researcher but that the employer can pick up in hiring (e.g., motivation, mission alignment) or by a direct impact of the treatment on effort on the job. While I cannot fully rule out that the treatment may motivate the new hires, I discuss some evidence against the effort channel in Section VC.

[16] New hires' allocation to teams is based on their regional preferences, team diversity, or slot availability, and it is orthogonal to treatment status. Seventy percent of new hires were sent to their preferred region. There are 51 communities in the sample, and the median team size is 4. In the pure control, 19 additional men were hired.

[17] The estimates in columns 1 and 2 of Table 3 are robust to winsorizing the grades at the ninety-fifth or ninety-ninth percentile and removing leavers, in order to correct for possible outliers (online Appendix Figure A.8, panel A). The coefficient on the "Lower Share" dummy is also robust to different imputation procedures for the grades of men who quit early, which are set to zero in the main tables.

TABLE 3—ON-THE-JOB OUTCOMES

	On-the-job performance			On-the-job attitudes			
	Grades semester 1 (1)	Grades semesters 2–3 (2)	Exit (3)	Perceived workload (4)	Perceived social impact (5)	Confidence in skills (6)	Intent to stay (7)
Panel A. Men							
Male Photo	−2.035	−7.440	0.007	0.127	0.189	0.165	0.155
	(2.603)	(4.510)	(0.108)	(0.117)	(0.100)	(0.109)	(0.090)
	[0.44]	[0.15]	[0.97]	[0.29]	[0.11]	[0.23]	[0.12]
	{0.67}	{0.31}	{0.96}	{0.23}	{0.18}	{0.16}	{0.14}
Lower Share	5.310	4.211	−0.072	−0.038	0.331	0.297	0.243
	(2.401)	(5.793)	(0.108)	(0.121)	(0.087)	(0.112)	(0.093)
	[0.03]	[0.51]	[0.49]	[0.77]	[0.00]	[0.01]	[0.01]
	{0.10}	{0.68}	{0.53}	{0.72}	{0.01}	{0.01}	{0.01}
Photo = info p	0.03	0.11	0.50	0.33	0.23	0.39	0.49
Mean DV	58.24	53.27	0.14	0.29	0.55	0.44	0.57
Mean DV in PC	55.89	45.49	0.32	0.41	0.67	0.72	0.89
Observations	43	43	43	97	67	113	89
Unique *Observations*	43	43	43	38	38	38	38
Panel B. Women							
Male Photo	1.323	−3.193	0.083	−0.042	−0.078	0.013	−0.003
	(1.033)	(2.150)	(0.043)	(0.055)	(0.053)	(0.056)	(0.044)
	[0.21]	[0.15]	[0.04]	[0.44]	[0.14]	[0.82]	[0.94]
	{0.20}	{0.25}	{0.14}	{0.72}	{0.35}	{0.94}	{0.93}
Lower Share	−0.851	−2.760	0.100	−0.050	−0.021	0.091	0.020
	(0.992)	(2.090)	(0.044)	(0.053)	(0.054)	(0.059)	(0.045)
	[0.40]	[0.19]	[0.02]	[0.33]	[0.70]	[0.14]	[0.66]
	{0.40}	{0.35}	{0.07}	{0.58}	{0.67}	{0.15}	{0.82}
Photo = info p	0.15	0.87	0.77	0.90	0.44	0.36	0.69
Mean DV	58.33	60.16	0.02	0.45	0.84	0.59	0.82
Mean DV in PC	59.74	57.83	0.07	0.33	0.84	0.61	0.90
Observations	191	191	191	480	297	495	403
Unique *Observations*	191	191	191	175	174	175	175
Gender differences							
Photo: men = women p	0.18	0.35	0.47	0.16	0.01	0.19	0.10
Info: men = women p	0.01	0.21	0.10	0.92	0.00	0.09	0.02

Notes: The table shows OLS estimates. The outcomes in columns 1 and 2 are weighted averages of on-the-job test scores. In the first semester there are six exams, including a first-month review (with higher weight within the semester). Three additional assessments are done in the second and third semesters. "Exit" is a dummy equal to one if the person left the program before completing it. "Perceived workload" is an indicator for perceiving the workload to be too high. "Perceived social impact" and "Confidence in skills" are indicators equal to one if a worker feels they are having a positive social impact at work, and for feeling confident in interacting with customers. "Intent to stay" is an indicator equal to one for being moderately or very likely to stay in the same community or job. "Unique *Observations*" is the number of unique respondents, and "*Observations*" is the number of observations. All the columns control for stratification variables and region fixed effects, and columns 4 to 7 control for survey wave fixed effects and cluster standard errors at the worker level.

Good performance on the job benefits both the employer and service beneficiaries but might come at the expense of workers' job satisfaction. Given the high turnover rates in social work, a concern is that new talented hires could become overloaded with cases and be more likely to leave. Table 3 provides evidence against these concerns. Men in the "Lower Share" group are not more likely to drop out of the program than those in the "Higher Share" (column 3). They are also significantly less likely to exit compared to the pure control ($p = 0.05$).

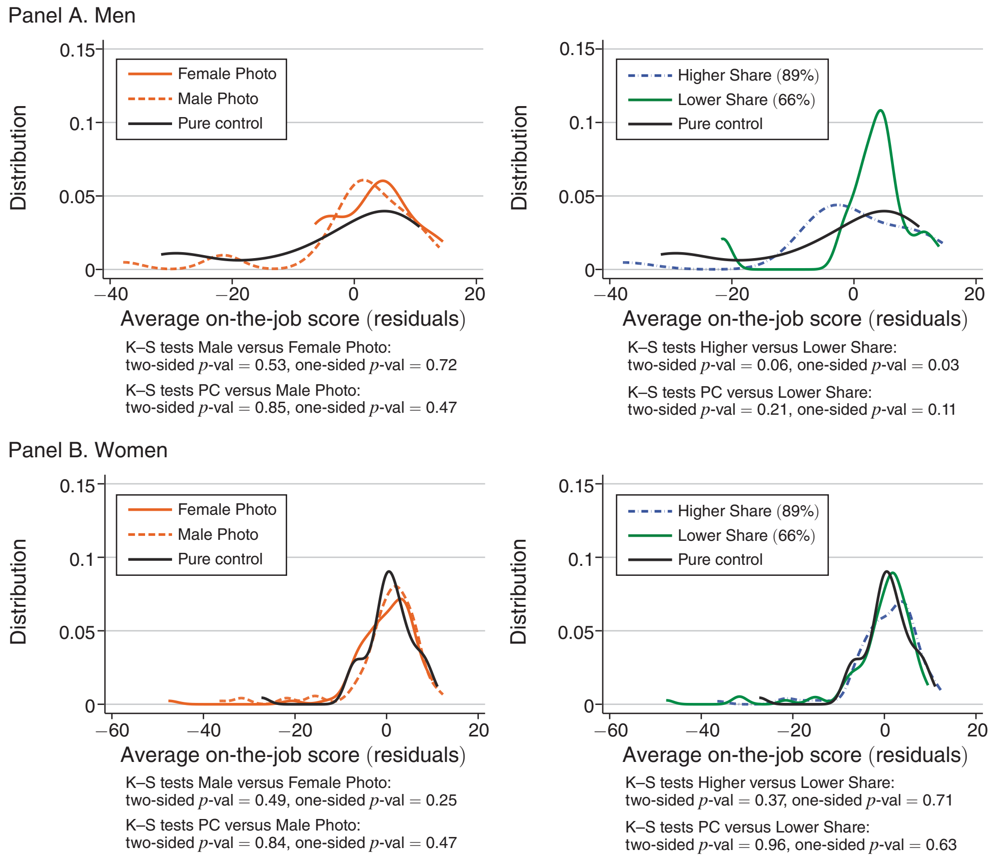

FIGURE 3. AVERAGE ON-THE-JOB TEST SCORES BY TREATMENT

Notes: The figure shows the distribution of men's (panel A) and women's (panel B) average on-the-job test scores, residualized after controlling for strata. "Average on-the-job score" is the weighted average of the scores in the nine performance assessments required in the job, where weights are given by the credits assigned to each test by the organization. The figure on the left-hand side shows the distributions by photograph treatment: the dashed line represents the "Male Photo," and the orange solid line represents the "Female Photo." The figure on the right-hand side shows the distributions by information treatment: the dashed line represents the "Higher Share" (89 percent), and the green solid line the "Lower Share" (66 percent). The black solid line represents the pure control.

Additional evidence about workers' job satisfaction comes from employee surveys, available for 89 percent of the new cohort of workers. When using these data, I keep repeated observations for each worker i and include survey-wave fixed effects in the main specification. As columns 4 and 7 of Table 3 show, men in the "Lower Share" do not have a higher perceived workload and are more likely to state that they want to keep being social workers in the future than in the "Higher Share." Online Appendix Table A.7 also shows no differences across treatment groups in men's concerns about the work environment. Furthermore, columns 5 and 6 of Table 3 show that men in the "Lower Share" group are more likely to be aware of the social impact of their work and to be confident in their skills when interacting with their beneficiaries.[18]

[18] Online Appendix Figure A.8, panel B shows that the coefficients of Table 3 are robust to controlling only for strata or adding controls for being sent to the preferred region and past application (which predicts assignment to

I find no effect of photographs on job performance. The top row of Table 3 shows that men in the "Male Photo" group tend to perform worse than in the "Female Photo" group, but none of the coefficients are statistically significant (see also online Appendix Figure A.8, panel A). Moreover, this gap is driven by the "Female Photo" outperforming the pure control. Although survey attitudes are slightly more positive among men in the "Male Photo" compared to the "Female Photo" group, online Appendix Figure A.8, panel B reveals that these coefficients are influenced by the choice of controls.

To summarize, disclosing information of a lower share of high-performing workers attracts a larger pool of male applicants, from which the employer selects talented hires who also perform better—and are more satisfied—once in local communities. While the male photograph has no impact on men's entry or quality, the female one attracts worse applicants who are substantially screened out by the employer, so that the few new hires in this group end up performing well.

B. *Women*

In this section, I discuss treatment effects on women's applications, hiring, and on-the-job outcomes. This analysis is crucial to know the net impact of the two manipulations, as employers can rarely target recruitment policies to only one gender.

Table 2 and online Appendix Figure A.3, panel B show that women are on average insensitive to the experimental information. The coefficient of $LowerShare_i$ on women's applications is -0.014 and statistically insignificant (column 1 of Table 2, panel B). I can reject the null hypothesis that the effect of the "Lower Share" on applications is the same for men and women ($p = 0.02$). Moreover, a regression pooling genders shows that the interaction between being a man and the "Lower Share" indicator remains significant and sizable when adding a large set of controls interacted with gender and/or the treatments (online Appendix Table A.9).

From this evidence, one could infer that informing potential applicants of a "Lower Share" of high performers is a silver bullet for the employer: it achieves higher diversity, better quality among the gender minority, and has no effect on the majority. However, I find that women react to photographs and, in particular, to the interaction between photographs and information.

Panel B of Table 2 shows that the "Male Photo" decreases women's applications by 5 percentage points compared to the "Female Photo." This represents 8.6 percent of the mean female application rate in the pure control, even if I cannot reject that the two genders react similarly to the photograph treatment ($p = 0.46$).[19] Most of the decline in applications comes from lower-ability women. For instance, the application rate is 63 percent in the "Female Photo" and drops to 55 percent in the "Male Photo" for women who did not study in a top-tier UK university, while there is no gap between the two arms for women from top universities.

a preferred location for men). The effects on survey outcomes are also not affected by controlling for on-the-job performance. A mediation analysis instead suggests that higher perceived impact and confidence mediate the information effect on performance.

[19] The effect is coming from both an increase in women's applications in the "Female Photo" and a decrease in the "Male Photo" treatment with respect to the pure control ($p = 0.26$ and 0.16, respectively. See online Appendix Table A.8).

Online Appendix Table A.8 shows that women's decrease in applications is entirely driven by the combination of a male photograph with the lower share information. Seeing a man in the advertisement has no impact on women's decisions when associated with the "Higher Share" but becomes demotivating when combined with the "Lower Share." This implies that the "Lower Share," despite its null average impact, may negatively affect women, depending on other contextual features such as counterstereotypical photographs.

Despite the negative impact of the "Male Photo" on women's application, on average, there is no impact of either photographs or information on women's offer and acceptance rates (columns 2 and 3 of Table 2), also compared with the pure control (online Appendix Table A.8). Thus, the decline in women's applications does not impact the quality of potential job offerees. Online Appendix Figures A.4, panel B and A.5, panel B and indeed show small and nonsignificant quality differences between treatments for women.

Table 3, panel B shows treatment effects for female hires ($observations = 191$ across treatment groups). While there is no impact of either treatment on performance, column 3 shows that women in both the "Male Photo" and the "Lower Share" groups are more likely to exit the program before completing it. This effect comes mainly from low-performing women, who have a below-median "index of desirable skills" or perform poorly in the first six months in the job (online Appendix Figure A.9).

To summarize, high-talent women remain largely unaffected by the policies tested to attract men into social work. In contrast, a male photograph—especially combined with the "Lower Share" treatment—discourages lower-ability women from entering or persisting in the job.

C. *Net Impact for the Employer*

Is disclosing information on a lower share of high-performing workers beneficial for the employer? If we only look at recruitment, the answer is positive. More and better men apply in the "Lower Share" group, while women's applications are not affected on average. However, when considering the outcomes for the incoming pool of workers, the desirability of the information policy becomes ambiguous. The reason is that information describing lower rates of success discourages low-performing women from staying in the job, creating a trade-off between performance and retention for the employer. Is the increase in performance among those women who remain in the job high enough to justify the loss of female workers?

To shed light on the way the employer evaluates this trade-off, I ran a survey with the management and recruitment team of my partner organization (see online Appendix Figure A.10). Respondents chose between two hypothetical cohorts of workers: one with no dropout but lower average performance and one with greater dropout but a resulting higher performance among stayers. The results show that 58 percent of respondents prefer a cohort with higher performance rather than longer retention, and this percentage increases to 71 percent if higher performance is associated with greater gender diversity. Overall, this evidence indicates that the net effect of disclosing the "Lower Share" information is positive, even after accounting for the cost of female turnover for the employer. Compared to the other experimental treatments, the "Lower Share" achieves the highest overall performance and workforce diversity (online Appendix Table A.10).

D. *Heterogeneity in Treatment Effects*

This section explores which types of men and women are most affected by the treatments in their application decision. Specifically, I consider heterogeneity in exposure to social work and outside options. The methodology used for this analysis is detailed in online Appendix B.

Heterogeneity by Exposure to Social Work: The impact of new information should be increasing in the initial uncertainty about the job. I build two empirical proxies for uncertainty: exposure to labor market gender segregation during teenage years and an index of familiarity with social work and the job offered by the partner firm.

For the former, I use microdata from the 2011 UK census (Office for National Statistics 2015) and construct the Duncan index of occupational gender segregation (Duncan and Duncan 1955) in the local area where a candidate went to secondary school and/or lived during teenage years.[20] The first two columns of Table 4 estimate heterogeneous treatment effects on applications, splitting the sample by exposure to higher- or lower-than-median Duncan index. The bottom row of panel A shows that men exposed to high occupational gender segregation react significantly more to new information, with their applications increasing by 13.5 percentage points in the "Lower Share" ($p = 0.02$). This effect represents 27 percent of the mean in the pure control, and it is different from the information effect in the lower segregation group ($p = 0.09$).

Columns 3 and 4 of Table 4 repeat the exercise, splitting the sample by higher- or lower-than-median familiarity with social work. The index averages baseline data on whether the person studied a common university subject for social workers, worked in health care or the public sector, attended any information event about the job, or applied in the past. The estimates indicate that the increase in applications in the "Lower Share" group is entirely driven by men with limited information about the job. Exposure to neither occupational gender segregation nor the job moderates their reactions to photographs.

Table 4, panel B shows heterogeneous treatment effects for women. Women's drop in applications is concentrated among those who are more familiar with the job. These women are 7.6 percentage points less likely to apply in the "Male Photo" group ($p = 0.0$), while the drop in applications from women who have less familiarity with the job is only 1.3 percentage points ($p = 0.65$). The effect of photographs and information is instead similar for women exposed to high or low occupational gender segregation.

[20]Exposure to gender segregation has been shown to affect the persistence of biased beliefs on group ability (Guryan and Charles 2013), an insight used by Arrow (1998) to explain long-term statistical discrimination. Online Appendix Table A.11 shows that men exposed to a high Duncan index indeed tend to have more uncertain beliefs about men's and women's performance in social work.

TABLE 4—HETEROGENEOUS TREATMENT EFFECTS BY BACKGROUND AND OUTSIDE OPTION

	DV: Applied and no dropout					
	By background				By outside option	
	Job genderization		Familiarity with job		Wage dispersion	
	Low (1)	High (2)	Low (3)	High (4)	Low (5)	High (6)
Panel A. Men						
Male Photo	−0.015	−0.014	−0.080	−0.005	−0.023	−0.018
	(0.050)	(0.052)	(0.057)	(0.044)	(0.045)	(0.056)
	[0.75]	[0.78]	[0.18]	[0.88]	[0.61]	[0.76]
	{0.99}	{0.99}	{0.89}	{0.97}	{0.98}	{0.96}
Lower Share	0.015	0.135	0.154	0.012	0.027	0.138
	(0.050)	(0.052)	(0.058)	(0.044)	(0.045)	(0.056)
	[0.77]	[0.02]	[0.01]	[0.79]	[0.55]	[0.01]
	{0.94}	{0.04}	{0.02}	{0.80}	{0.83}	{0.04}
Photo = info p	0.66	0.05	0.00	0.79	0.44	0.05
Mean DV	0.59	0.45	0.44	0.57	0.56	0.45
Mean DV in PC	0.57	0.51	0.46	0.57	0.58	0.46
Observations	400	379	297	510	483	324
Photo: high = low p	0.99		0.30		0.95	
Info: high = low p	0.09		0.05		0.12	
Panel B. Women						
Male Photo	−0.071	−0.036	−0.013	−0.076	−0.060	−0.024
	(0.024)	(0.024)	(0.028)	(0.020)	(0.019)	(0.034)
	[0.01]	[0.11]	[0.65]	[0.00]	[0.00]	[0.47]
	{0.01}	{0.30}	{0.62}	{0.00}	{0.01}	{0.45}
Lower Share	−0.016	−0.013	−0.045	−0.001	−0.020	−0.007
	(0.024)	(0.024)	(0.028)	(0.020)	(0.019)	(0.034)
	[0.51]	[0.59]	[0.14]	[0.94]	[0.29]	[0.84]
	{0.89}	{0.89}	{0.32}	{1.00}	{0.76}	{0.99}
Photo = info p	0.10	0.48	0.43	0.01	0.14	0.71
Mean DV	0.62	0.60	0.55	0.63	0.61	0.57
Mean DV in PC	0.57	0.61	0.56	0.61	0.60	0.55
Observations	1,710	1,693	1,240	2,273	2,644	869
Photo: high = low p	0.30		0.07		0.36	
Info: high = low p	0.93		0.20		0.73	
Gender differences						
Photo: men = women p	0.30	0.69	0.29	0.14	0.45	0.93
Info: men = women p	0.56	0.01	0.00	0.79	0.34	0.03

Notes: The table shows OLS estimates with controls chosen using PDS Lasso. Columns 1 and 2 split the sample by the Duncan index of occupational gender segregation in the local area where a candidate went to secondary school or currently lives. Columns 3 and 4 split the sample by an index of familiarity with social work and the job offered by the partner organization. In columns 5 and 6, the sample is split by the wage dispersion faced by participants in their outside option in the United Kingdom. In all the columns, levels "high" or "low" are defined for values of the indexes above or below the gender-specific median in the experimental sample. See online Appendix B for details on the construction of the indexes. The rows "Photo: high = low p" and "Info: high = low p" report the p-value of a test of equality of the coefficients on the "Male Photo" and "Lower Share," respectively, for the samples with high versus low levels of the variable used for the sample split.

Heterogeneity by Outside Option: In the "Lower Share" group, the increase in male applications without a decline in quality is consistent with a model of negative sorting in social work, where the higher-ability marginal applicant faces higher returns to talent in his outside option compared to the average applicant (Heckman and Taber 2010). In what follows, I empirically test this hypothesis.

I use as a proxy for returns to talent the average wage dispersion faced in the UK labor market according to a person's field of study. For an individual who studied subject s, wage dispersion is calculated as the average $75/25$ interquartile range of wages across industries, weighted by the proportion of graduates in s working in each industry. Columns 5 and 6 of Table 4 show heterogeneous treatment effects splitting the sample by above- or below-median wage dispersion. As hypothesized, the increase in applications in the "Lower Share" group is stronger among men facing above-median wage dispersion. While I cannot reject the equality of coefficients between the two subsamples at conventional levels $(p = 0.12)$, the coefficient on the treatment dummy is more than four times larger for candidates with steeper outside options. Accordingly, online Appendix Table A.13 further shows that the "Lower Share" effect is qualitatively stronger among male candidates with a higher predicted hourly wage and job performance. This lines up with the hypothesis that the marginal male applicant induced to apply by the information treatment faces better returns to talent in their outside option, consistent with the increase in quality among top-ranked applicants. Table 4 and online Appendix Table A.13 instead confirm that low-ability, flat–outside option women are disproportionately discouraged by the "Male Photo."

V. Interpretation

Why are photographs ineffective, while the "Lower Share" information attracts more (and better) men into social work? To answer this question, I use auxiliary survey experiments to understand how participants interpreted the experimental messages. Drawing on these survey data, in the following subsections I discuss the primary interpretations of the two treatments and then present the main alternative explanations. Online Appendix F rules out less relevant interpretations.

Survey Experiments.—I conduct the auxiliary survey experiments using a between-subject design. After being assigned to one of the treatment emails used in the field experiment and mandatory comprehension checks, respondents answered questions regarding characteristics of the advertised job, its applicants, and its workers.[21] I collected these survey data from two out-of-trial samples. The first sample consists of applicants for the job offered by the partner organization in the following year (2018) and tests reactions to the treatments within a subject pool as similar as possible to that of the RCT $(observations = 242)$. The second sample was recruited through the platform Prolific and includes online students and workers, matched on observable characteristics with the field participants $(observations = 262)$.

By using out-of-trial samples, I overcome potential risks with directly questioning RCT participants about the email contents. A survey within the RCT sample could have indeed affected reactions to the treatment (e.g., making gender salient) and thus interfered with the double-blind design. Moreover, including the Prolific

[21] Online Appendix C describes the survey implementation and questions used. For all the survey variables, online Appendix Table A.14 shows descriptive statistics and MHT-adjusted p-values of mean differences between treatment groups, by gender. Online Appendix Table A.15 shows the equivalent information, by survey sample. The analysis in this section was pre-registered except for the right-hand side of Figure 5, which originated from conversations with the partner organization after registration.

sample allows me to investigate whether survey results generalize beyond just the candidates interested in the job. Online Appendix Table A.15 shows that the two samples reply similarly to most survey questions, thereby supporting the generalizability of the main interpretations that follow.

A. *Photographs Interpretation*

The purpose of the photograph manipulation was to explore whether gender identity concerns deter men from applying to female-dominated jobs. Figure 4 presents survey responses regarding this aspect. Overall, the evidence indicates that the experimental photographs generated variation in respondents' perceived gender shares and beliefs about the employer's gender preferences. However, they did not influence men's perceptions regarding the gender stereotype associated with social work.

The first set of bars in the figure illustrates participants' beliefs about the female share of applicants for the job.[22] When shown the female photograph, respondents report an average female share of 73.8 percent among applicants, which decreases to 68.3 percent when shown the male photograph ($p = 0.00$). This gap in shares is almost identical for male (panel A) and female respondents (panel B).

The second and third variables displayed in the figure show that compared to the "Female Photo," in the "Male Photo" group, men are 44 percentage points more likely to believe that the employer wants to attract men ($p = 0.00$). They are also 18 percentage points less likely to perceive gender or race discrimination from customers ($p = 0.02$). While women's updating regarding the employer's gender preferences is even stronger relative to men ($p = 0.00$), their views on customers' discrimination do not change in the two treatment groups.

The last three sets of bars in Figure 4 relate to respondents' explicit or implicit association between social work and women. In the "Female Photo" group, 58 percent of men believe the job is desirable for a man, while 97 percent think it is desirable for a woman. These numbers increase and decrease by 7 percentage points, respectively, in the "Male Photo," but the differences are marginally statistically significant or not at all. In contrast, women significantly update their beliefs on both questions, with gaps between the two treatment arms exceeding 13 percentage points ($p < 0.01$). Similar trends emerge when examining the proportion of respondents with an above-median implicit association between social care and women, measured through a single-target IAT. Photographs do not alter men's IAT scores, but they do result in a change for women.[23]

To summarize, when seeing a male photograph, both genders expect more male applicants and for the employer to be more willing to hire men (both measures are robust to MHT corrections; see online Appendix Table A.14). However, the finding that the association between women and social work is sticky for men—but not for women—might help explain why using male photographs has no effect on men's entry into the job, while it has a negative effect on women's applications.

[22] Priors on gender shares lie between the posteriors in the two photograph arms (online Appendix Figure A.11, panel A). Additional surveys show that respondents reply similarly when asked about workers' gender composition rather than applicants.

[23] The IAT was an optional task, and 57 percent of respondents took part in it (47 percent and 68 percent in the applicant sample and the Prolific sample, respectively).

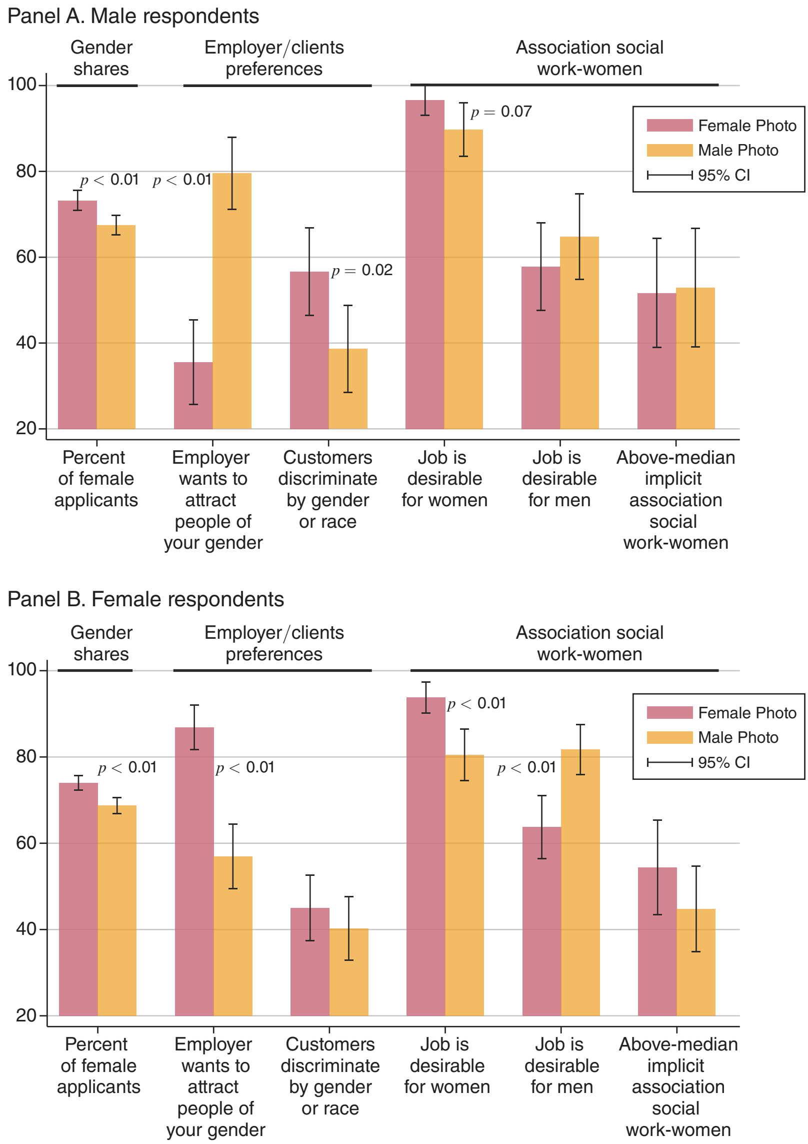

FIGURE 4. OUT-OF-TRIAL SURVEYS: INTERPRETATION OF THE PHOTOGRAPH TREATMENT

Notes: Panels A and B show responses from the out-of-trial surveys, by photograph treatment and gender. "Percent of female applicants" is the believed share of female applicants. "Employer wants to attract people of your gender," "Job is desirable for men," "Job is desirable for women," and "Customers discriminate by gender or race" show the share of people who agree with the given statement. The last variable is constructed from a single-target IAT between social care and women. Gender differences: treatment effects differ by gender in whether the employer wants to attract men ($p = 0.00$) and in an index of job-women association, which includes the IAT and the two questions on job desirability by gender ($p = 0.02$).

B. *Information Interpretation*

The information treatment was designed to address men's expectations of success, which could be pessimistic or uncertain due to limited exposure to female-dominated jobs. The intervention may affect these expectations directly (by changing men's expected job performance) or indirectly (by influencing beliefs about workplace features that in turn affect success). The survey evidence discussed in the following paragraphs suggests that the randomized information did change the expected premium for talent in the job, thus influencing the degree of success that high-ability respondents could achieve compared to those with lower ability.

To arrive at this preferred interpretation, the left-hand side (LHS) of Figure 5 displays the average response to several survey questions, by information treatment and gender.[24] The initial six variables presented in Figure 5, panel A show that the experimental information has no discernible impact on men's perceptions of the difficulty of job tasks, promotion prospects, or quality standards set by the employer. Similarly, men in the two information treatments share similar beliefs on the effort required to apply, the number of applicants, and how the employer would rank them for the job out of 100 applicants.

The last three variables in the LHS of Figure 5 pertain to beliefs about success in the advertised social work role. The information treatment does not affect men's self-reported ability, measured as their expected performance in service delivery. Nonetheless, men in the "Lower Share" group predict a lower ranking for themselves in the job (a decrease of 6.5 percentage points, $p = 0.09$) and report that fewer applicants have the potential for high performance (66 percent versus 75 percent, $p < 0.01$ and robust to MHT) compared to men in the "Higher Share" group.

To see how these patterns may reconcile with my main experimental results, the right-hand side (RHS) of Figure 5, panel A shows treatment effects on male respondents' expected ranking for the job, splitting the sample by self-reported ability above/below men's median. In the "Higher Share" treatment, ability differences do not map into different rankings for the job, as men of different self-reported abilities expect similar rankings ($p = 0.17$). In contrast, low-ability men expect a significantly lower ranking than high-ability men in the "Lower Share" group ($p = 0.015$).[25] Notably, this difference in expected ranking within the "Lower Share" group stems from men with below-median ability revising their expectations downward, while men with above-median ability anticipate similar rankings in the two information groups.

Figure 5, panel B reveals that for most of the variables discussed so far, women's beliefs closely align with those of men. The only differences are that women perceive the job as more difficult in the "Lower Share" group relative to the "Higher Share" ($p = 0.00$). However, they also expect to perform better (by 4.9 percentage

[24] Online Appendix Table A.14 shows that the information treatment does not affect beliefs targeted by the photograph treatment (discussed in Section VA) and vice versa. Online Appendix Figure A.13 further shows no significant interaction between two manipulations on the main elicited beliefs.

[25] The results are robust to using different thresholds of self-reported ability, when using on-the-job predicted performance based on observables instead of the self-reported measure (see online Appendix Table A.16) and when winsorizing the ranking variable to adjust for possible outliers. Qualitatively, I also obtain the same results when I use the average between one's own ranking and the employer's ranking, or a winsorized version of the latter.

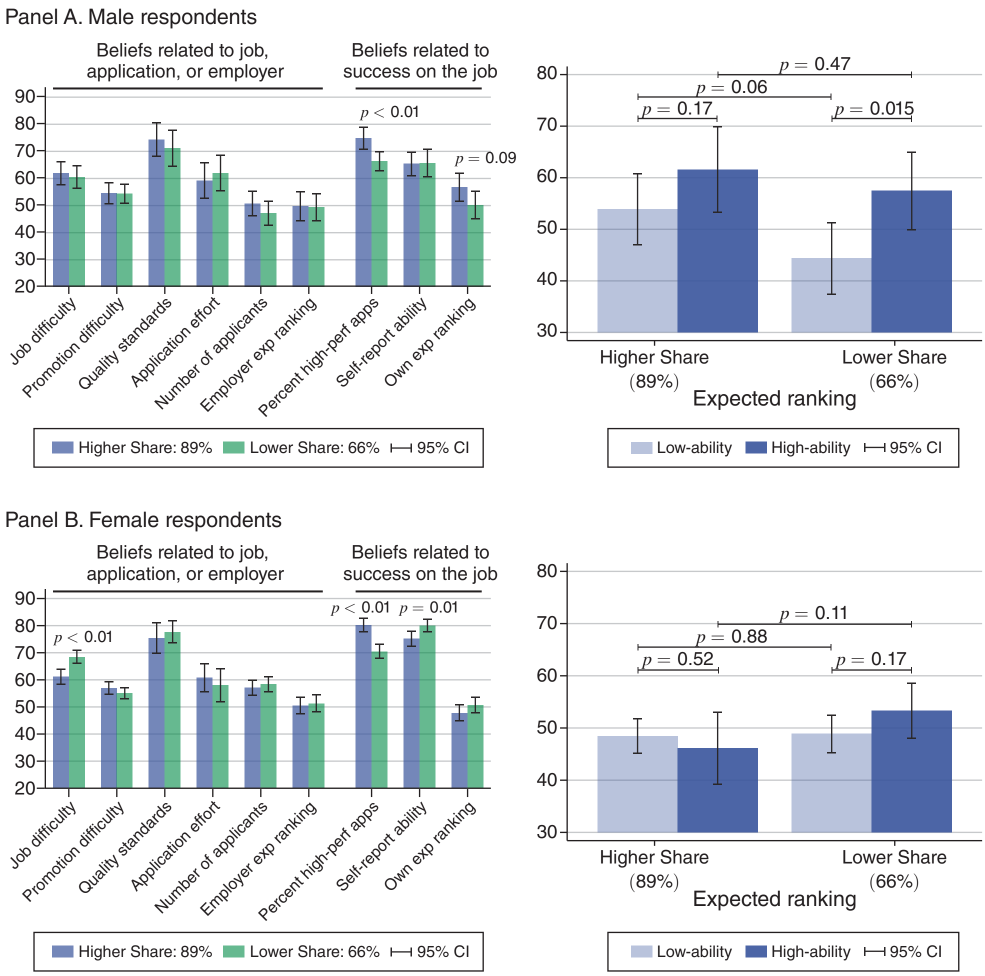

FIGURE 5. OUT-OF-TRIAL SURVEYS: INTERPRETATION OF THE INFORMATION TREATMENT

Notes: Panels A and B show responses from the out-of-trial surveys, by information treatment and gender. Shown on the left-hand side of the figure, "Job difficulty," "Promotion difficulty," "Quality standards," and "Application effort" asked participants to rate this job feature from 0 to 100. "Percent high-perf apps" indicate beliefs on the share of applicants with the potential to be high performers on the job. "Self-report ability" is the respondents' expected performance in interacting with customers. "Own exp ranking" and "Employer exp ranking" indicate the ranking that the respondent expects for the job or that they expect from the employer. The right-hand side of the figure splits the sample below/above the gender-specific median self-reported ability and shows the average "Own exp ranking" within each subsample. Gender differences: treatment effects differ by gender in job difficulty ($p = 0.02$) and one's own expected ranking ($p = 0.03$).

points, $p = 0.01$) and do not change their expected ranking in the former group compared to the latter. In line with the findings for men, the RHS of Figure 5, panel B shows that the gap in expected ranking between high- and low-ability women is also larger in the "Lower Share" group compared to the "Higher Share" but is not statistically significant.

In summary, Figure 5, panel A shows that compared to those in the "Higher Share" treatment, men in the "Lower Share" anticipate lower potential success among applicants and a higher premium for ability in relative job evaluations. This evidence provides three possible explanations for men's increased entry in the "Lower Share" treatment. First, men may be drawn to a more competitive work environment, in line with existing economic literature and gender differences in response to information (Niederle and Vesterlund 2007).[26] Second, the "Lower Share" information may signal that the employer's evaluation more accurately reflects talent. Instead of giving similar grades to everyone—as in the "Higher Share"—the employer is careful in ranking high-performing workers better than low-performing ones. Third, candidates may be attracted by the opportunity to make a positive impact on the job and improve service delivery compared to the past. While both the 89 percent and 66 percent statistics indicate a competent workforce, the latter case implies a greater potential for talented hires to make a difference in their roles.

These three leading interpretations find support in both qualitative interviews and additional surveys. In-depth interviews with RCT participants particularly support the idea that the "Lower Share" encouraged those motivated by the prospect of improving outcomes for beneficiaries or the employer.[27] The three main interpretations of increased competitiveness, evaluation accuracy, and potential effectiveness also emerge in an additional survey employing open-ended questions about the meaning of the experimental information (see online Appendix E).

It is important to note that according to Figure 5, panel A, high-ability men's expected job ranking remains the same in both information treatments despite their increased likelihood of applying in the field experiment. One explanation may be that the ability proxies used in the surveys are too coarse to detect nuanced belief changes among individuals at the upper end of the talent distribution (those more likely to apply as an effect of the treatment in the RCT). Alternatively, high-ability candidates may change their views of the employer rather than their self-assessment. Their motivation to apply in the "Lower Share" group may arise from the belief that their evaluations will more accurately reflect their skills or that their talent will yield a higher marginal benefit for the employer. Interviews with RCT participants align with these hypotheses, further corroborated by online Appendix Table A.16 and online Appendix Figure A.12, which indicate that men in the top decile of predicted job performance ($observations = 21$) positively revise their expectations of their own and the employer's ranking in the "Lower Share."

[26] However, men's higher willingness to compete is especially found in masculine tasks (Dreber, von Essen, and Ranehill 2014; Flory, Leibbrandt, and List 2015). Moreover, many of the participants come from prosocial organizations or humanistic subjects, which negatively correlate with competitiveness (Buser, Niederle, and Oosterbeek forthcoming; Non et al. 2022).

[27] For instance, one interviewee said, "66 percent is good, it makes you want to feel part of it. I am not sure about my performance, but I could say: we can do better than it!" Another said, "66 percent is high, but not super high. It is something to be pushed." Conversations with social workers and managers from the partner firm also highlight that applicants are motivated by the drive to make a difference and the feeling that their talent is needed. Furthermore, text analysis of open-ended questions in the application form indicates an increase of words related to effectiveness and making a difference for men in the "Lower Share" group.

C. *Alternative Interpretations*

The intervention may affect other beliefs not targeted by design. Moreover, the treatments may have affected outcomes in ways not related to worker's self-selection into application, for example, influencing the employer's screening and applicants' effort.

Does the Information Treatment Change Hiring Expectations? Survey respondents believe that applicants are less likely to succeed in the job in the "Lower Share" group compared to the "Higher Share." This evidence may indicate that they expect worse competitors and thus higher chances of being hired in the "Lower Share" treatment. However, this explanation is at odds with the main empirical findings, as it predicts that low-skilled individuals should shy away from better competitors in the "Higher Share" group, which should have higher quality than the other information treatment (Lazear, Shaw, and Stanton 2018; Wright et al. 2019).

Does Any Treatment Signal Higher Wages or Social Status? Online Appendix Table A.14 shows that neither photographs nor information conveys signals about the wage level or the social status of the job. The only exception is women's belief that the job has lower social status when receiving the "Lower Share" compared to the "Higher Share" information (not robust to MHT corrections).

Are Performance Outcomes Caused by Changes in Employer's Screening? To attribute changes in performance outcomes to the causal effect of the treatment on applicants' composition, one needs to rule out the possibility that the treatment affects the employer's screening criteria (Ashraf et al. 2020). To check for this possibility, I use the following model:

$$o_i = \sum_j \alpha_j^{T^1} T_i^1 X_i^j + \sum_j \alpha_j^{T^2} T_i^2 X_i^j + \mathbf{S}_i' \lambda + \epsilon_i,$$

where o_i is one if applicant i got a job offer, T_i^1 and T_i^2 are indicator variables for the two arms in each condition (e.g., male/female photographs), $\mathbf{S}_i$ are strata controls, and X_i^j are indicators for having a certain qualification (e.g., a first grade). Online Appendix Table A.17 shows that observable characteristics are equally predictive of getting an offer across treatments. Thus, the employer's selection criteria are not affected by treatment assignment, as expected in a double-blind design.

Are Performance Outcomes Caused by Changes in Effort? The higher quality of male applicants and workers is consistent with better selection generated by the "Lower Share" treatment. An alternative explanation may be related to effort. That is, chances to make a difference or stand out motivate men to put more effort into the hiring process or their on-the-job performance. While it is not possible to completely dismiss this potential explanation, two lines of evidence suggest otherwise. First, any motivating effect of the treatment should be stronger right after receiving the invitation-to-apply email. In contrast, online Appendix Table A.18 shows that men in the two information treatments do not differ in the effort put into the application (e.g., the share of fields filled in or the length of responses). Second, the small sample of new male hires report the same actual and perceived workload in the two arms (Table 3 and online Appendix Table A.7).[28]

VI. Discussion

In the field experiment, gendered photographs were ineffective in attracting men to social work, while information of lower past success encouraged more male applications. There are a range of possible reasons for, and implications of, these findings for employers in pink-collar professions.

Why Is the Male Photograph Ineffective? The null effect of the male photograph on men's applications speaks to policies advocating that media campaigns portraying male workers will help attract men to teaching or nursing roles (Flynn 2006; *Economist* 2018). For instance, in 2002 the Oregon Center for Nursing tried to appeal to young men by launching the notorious "Are you man enough to be a nurse?" advertisement, which portrayed a lineup of masculine male nurses engaged in extreme sports. A realistic representation of male nurses is a pillar of the biggest recruitment drive in the NHS history (see McGonagle 2019). While some sources indicate that this campaign drove a 9 percent increase in men's enrollment in nursing from 2018 to 2019 (NHS 2019), my findings prompt caution in considering this a causal link.[29]

One way to reconcile my results with these policies is through men's selection into the experimental sample. If participants care less about gender-related factors than the average man, the estimated effect of the male photograph will be a lower bound. I can address this channel through an additional experiment that I ran with the same partner organization (see online Appendix D). I tested whether advertisements portraying men could be effective in encouraging a wider population of male students—not yet interested in or aware of social work vacancies—to apply for the job. In this broader population, I find no evidence that male photographs increase male applications. This result extends the external validity of the main finding in the paper.

A plausible explanation for the null effect of male photographs emerges from my auxiliary surveys. Even if portraying a man in recruitment messages increases men's perceptions of the proportion of male applicants, the treatment is not powerful enough to change men's association—implicit or explicit—between social work and women. Perhaps advertisements that induce larger changes in perceived male shares (e.g., portray a group of men) may be successful in affecting this relationship. However, it is hard to generate such change under the constraint of moving beliefs in realistic ranges that do not create false expectations for the recipients.

[28] In surveys, the effects of information often decay rapidly (Coppock 2016). This makes it unlikely that the "Lower Share" affected effort only when people started the job many months after the experiment. I also find no evidence that men's scores come from a "surprise" when comparing peers' expected and actual performance.

[29] See https://www.england.nhs.uk/2019/02/young-male-nursing-applicants-surge-after-we-are-the-nhs-recruitment-campaign/; https://www.theguardian.com/society/2019/feb/24/nhs-breakthrough-more-men-study-nursing.

Why Does the "Lower Share" Information Attract More Men? As discussed in Section VB, the "Lower Share" appears to communicate that talented workers can achieve better evaluations on the job compared to untalented ones. The positive impact of this treatment on application rates indicates that the information provided effectively addresses men's uncertainty about talent recognition in female-dominated fields. This aligns with the evidence presented in online Appendix Figure A.2, panel C, which shows that men underestimate men's returns to skills in female-dominated jobs, either when compared to women in these jobs or both genders in male-dominated jobs. Given these underlying priors, messages emphasizing that the employer cares about rewarding talent can be an effective strategy for enhancing workforce diversity and quality in female-dominated sectors.

A natural question is whether the exact types of reward to talent advertised by the employer matter. In my experiment, participants could see returns to ability as either being related to career incentives (e.g., vertical progression or other forms of organizational recognition) or to prosocial incentives (the positive social impact that talent can have on the beneficiaries). When the employer does not favor one motivation over the other, this ambiguity in recruitment messages can be beneficial. The "Lower Share," by not specifying the exact meaning of returns to talent, may appeal to both career-oriented and socially motivated individuals. Therefore, a similar impact can be expected in other applicant pools where both career and prosocial motivations are present, such as in public sector jobs (Dal Bó, Finan, and Rossi 2013; Ashraf et al. 2020).

How Big Is the Impact of Information? As a first benchmark, I compare my estimates to studies that employ similar light-touch interventions to encourage women's entry in tech or corporate jobs. The effect of the "Lower Share" treatment on men's applications—of 11 percent compared to the pure control and 14 percent compared to the "Higher Share"—falls within the range of effects observed in three recent experiments. In a study investigating the provision of information about role models, Del Carpio and Guadalupe (2022) find a treatment effect on women's applications between 19 and 100 percent depending on the exact informational bundle tested. Changing the language on required qualifications in job vacancies, Abraham, Hallermeier, and Stein (2022) find an increase in applications by women (and men) of 7 percent. Gee (2019) finds that providing information about the number of people who had already applied to a position increased further applications by 3.5 percent. Differences in magnitude can be explained by different application costs or base-rate levels of applications across setting.

As a second benchmark, studies that exogenously varied wage levels in clerical or public sector jobs find effects on application rates between 26 percent and 45 percent (Abebe, Caria, and Ortiz-Ospina 2021; Dal Bó, Finan, and Rossi 2013). Compared to the "Higher Share" information, the effect of the "Lower Share" is between one-third and one-half of these previous estimates, but it is also nearly costless for the employer. Using both benchmarks, the "Lower Share" therefore appears to be a cost-effective way for employers to meaningfully change application behavior by men.

Why Do Men and Women React Differently to Information? The "Lower Share" discourages (low-ability) women's applications, but this effect is only observed when combined with a male photo. While it is beyond the scope of this paper to fully explain women's behavior, preferences or beliefs can help unravel these gaps. For instance, women may dislike working with men in challenging environments (Niederle and Vesterlund 2007), a channel that aligns with evidence that the differential effect of information on men and women survives a high-dimensional set of controls interacted with gender (online Appendix Table A.9).

Alternatively, photographs may interact with information to shape women's beliefs about their fit within the job (Croson and Gneezy 2009; Dreber, von Essen, and Ranehill 2014; Coffman, Collis, and Kulkarni 2023a). Unlike men, female survey respondents perceive the job to be more difficult in the "Lower Share" group (gender difference $p = 0.02$). Moreover, within this group, women perceive that the employer's standards are higher in the "Male Photo" compared to the "Female Photo" (by 8 percentage points, $p = 0.07$). Thus, the male photograph seems to intensify women's perception of difficulty in the "Lower Share" treatment, with a negative impact on low-ability women (who may have thought to be a good fit for the job because of their gender). In line with this channel, I find that turnover is highest for women working in teams with a high male share and who perform worse than their peers (online Appendix Figure A.9, panel B). An explanation based on preferences would predict higher female turnover in teams with more men, independent of one's own performance.

VII. Conclusion

The decline of blue-collar employment across the developed world challenges men's traditional roles in society by creating idleness and financial insecurity (Autor, Dorn, and Hanson 2013), particularly for lower-skilled individuals (Autor, Dorn, and Hanson 2019; Coile and Duggan 2019). In contrast, female-dominated sectors like health care and education are expanding, with relatively minor automation risks (Nedelkoska and Quintini 2018; Cortes and Pan 2019), yet their male workforce has hardly changed over time. Understanding the interplay of gender norms and gender-specific information in evolving labor markets is crucial for unlocking new opportunities for men transitioning from declining sectors (Reeves 2022).

This paper provides a proof of concept that limited information, rather than gender-related factors, can explain the narrow entry of men into female-dominated jobs. Increasing the expected premium for talent in the job increases men's applications in social work, especially when their exposure to the sector is limited, and improves selection on talent. At the same time, information that is motivating for men can discourage women, indicating that the way in which men are recruited might affect the prevailing gender group in these occupations.

Historically, advertising has been a common strategy to change gender shares in male-dominated occupations. Rosie the Riveter is a long-lived testimonial of the crucial role of advertising in recruiting women to supply-short male jobs during WWII (Honey 1985). This legacy inspired recent attempts to attract men to female-dominated sectors by portraying male nurses or teachers in advertisements. My results caution against strategies designed to promote a new male identity in these roles without addressing informational constraints.

Nevertheless, one should be careful in extrapolating my results into other contexts or samples. On the one hand, candidates in my experiment may have worse outside options or pay more attention to recruitment contents. On the other hand, they may have less malleable priors or weaker gender identity concerns. Extensions of this work for future research may include conducting similar experiments on broader sets of applicants in other industries or countries, where gendered norms around work are even stronger.

While this paper focuses on the talent-diversity trade-off, gender integration in female-dominated jobs may bring about challenges or societal costs. First, diversity should not be mistaken for inclusion. Although I do not find evidence of this issue in my setting, harassment and discrimination against men are not uncommon in workplaces with a high female share (Folke and Rickne 2022). Inclusion initiatives may be important complements of diversity recruitment.

Second, integration along one margin may create inequality in others. Pink-collar jobs have a reputation of being "pink ghettos," as they usually offer low salaries and few career opportunities, are considered of lower social status, and have reduced worker satisfaction. Keeping fixed the current incentive structure, gender integration in these jobs may either equalize or amplify disadvantages. Men's entry may facilitate the reallocation of women to more productive matches, and depending on the individual counterfactual, it may also help to avoid male idleness. But I also find that the information treatment disproportionately attracts nonwhite men into social work, implying that ethnic and racial minorities may get trapped into yet another low-status, low-income sector, even if across the traditional gender barriers. At the same time, nonwhite workers may have positive downstream impacts on beneficiaries who share similar demographics. These are important issues for further research.

Finally, what would happen to female jobs if men entered them in large numbers? My paper makes progress on this question by showing that low-ability women might be discouraged from staying in these jobs. However, the long-term labor consequences for these women remain uncertain. Would the gender composition ultimately tip in favor of men (Pan 2015; Ensmenger 2012)? Or could there be backlash by female incumbents (Rudman and Fairchild 2004)? These are important areas for future research. Hopefully tackling these themes can help communities, such as those in the Rust Belt or northern England, face the challenges of a rapidly evolving economy.

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